MATLAB Scripts – Sets 1 and 2

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Set 1:

Given the vectors $\boldsymbol{R_1} = \boldsymbol{a_x} + 2\boldsymbol{a_y} + 3\boldsymbol{a_z}$, $\boldsymbol{R_2} = 3\boldsymbol{a_x} + 2\boldsymbol{a_y} + \boldsymbol{a_z}$. Find a) the dot product $\boldsymbol{R_1} \cdot \boldsymbol{R_2}$, b) the projection of $\boldsymbol{R_1}$ on $\boldsymbol{R_2}$, c) the angle between $\boldsymbol{R_1}$ and $\boldsymbol{R_2}$. Write a MATLAB program to verify your answer.

Analytical Result:

a) Find the dot product $R_1 \cdot R_2$:

$$R_{1x} = 1, R_{1y} = 2, R_{1z} = 3$$

$$R_{2x} = 3, R_{2y} = 2, R_{2z} = 1$$

$$R_1 \cdot R_2 = (R_{1x} \times R_{2x}) + \left(R_{1y} \times R_{2y}\right) + (R_{1z} \times R_{2z})$$

$$R_1 \cdot R_2 = (1 \times 3) + (2 \times 2) + (3 \times 1)$$

$$R_1 \cdot R_2 = 3 + 4 + 3$$

$$R_1 \cdot R_2 = 10$$

$$\therefore \textit{the dot product } R_1 \cdot R_2, \textit{is equal to } 10 \textit{ which was also the result the MATLAB code generated meaning that the answer is verified}$$

b) Find the projection of R_1 on R_2:

$$\|R_2\| = \sqrt{{R_{2x}}^2 + {R_{2y}}^2 + {R_{2z}}^2} = \sqrt{3^2 + 2^2 + 1^2} = \sqrt{9 + 4 + 1} = \sqrt{14}$$

$$\|R_2\|^2 = (\sqrt{14})^2 = 14$$

$$R_1 \cdot R_2 = 10$$

$$proj_{R_2}R_1 = \frac{R_1 \cdot R_2}{R_2 \cdot R_2}R_2 = \frac{R_1 \cdot R_2}{\|R_2\|^2}R_2$$

$$proj_{R_2}R_1 = \frac{10}{14}(3a_x + 2a_y + a_z)$$

$$proj_{R_2}R_1 = \frac{5}{7}(3a_x + 2a_y + a_z)$$

$$proj_{R_2}R_1 = \frac{15}{7}a_x + \frac{10}{7}a_y + \frac{5}{7}a_z$$

$$proj_{R_2}R_1 \approx 2.1429a_x + 1.4286a_y + 0.7143a_z$$

$$\therefore \textit{the projection of } R_1 \textit{ on } R_2 \textit{ is equal to } \frac{15}{7}a_x + \frac{10}{7}a_y + \frac{5}{7}a_z \textit{ or approximately } 2.1429a_x + 1.4286a_y + 0.7143a_z \textit{ which was also the result the MATLAB code generated meaning that the answer is verified}$$

c) Find the angle between R_1 and R_2:

$$\|R_1\| = \sqrt{{R_{1x}}^2 + {R_{1y}}^2 + {R_{1z}}^2} = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{1+4+9} = \sqrt{14}$$

$$\|R_2\| = \sqrt{{R_{2x}}^2 + {R_{2y}}^2 + {R_{2z}}^2} = \sqrt{3^2 + 2^2 + 1^2} = \sqrt{9+4+1} = \sqrt{14}$$

$$R_1 \cdot R_2 = 10$$

$$R_1 \cdot R_2 = \cos\theta \, \|R_1\|\|R_2\|$$

$$\cos\theta = \frac{R_1 \cdot R_2}{\|R_1\|\|R_2\|}$$

$$\cos\theta = \frac{10}{\sqrt{14}\sqrt{14}}$$

$$\cos\theta = \frac{10}{14} = \frac{5}{7}$$

$$\theta = \cos^{-1}(\frac{5}{7})$$

$$\theta \approx 44.4°$$

$\therefore$ *the angle between* R_1 *and* R_2 *is approximately to* $44.4°$ *which was also the result the MATLAB code generated meaning that the answer is verified*

MATLAB Output to Verify Answers:

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Command Window
a) The dot product between R1 and R2 is 10.000000
b) The projection of R1 on R2 is [2.1429  1.4286  0.7143]
c) The angle between R1 and R2 in radians is 0.775193
c) The angle between R1 and R2 in degrees is 44.415309
fx >>
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Set 2:

The surfaces $r = 0$ and $r = 2$, $\phi = 45°$, $\phi = 90°$, $\theta = 45°$ and $\theta = 90°$ define a closed surface. Find the enclosed volume and the area of the closed surface S. Write a MATLAB program to verify your answer.

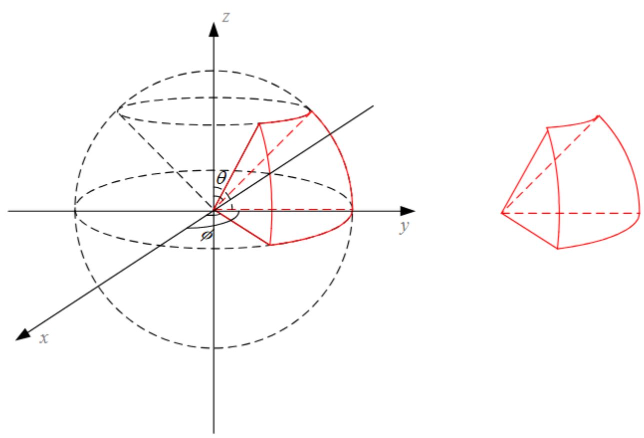

Figure 2.6 The surface of the exercise of Set 2.

Analytical Result:

1) Find the enclosed volume of the closed surface:

$$0 \le r \le 2, \frac{\pi}{4}(45°) \le \phi \le \frac{\pi}{2}(90°), \frac{\pi}{4}(45°) \le \theta \le \frac{\pi}{2}(90°)$$

$$dV = r^2 \sin\theta \, dr d\theta d\phi$$

$$V = \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{r=0}^{2} r^2 \sin\theta \, dr d\theta d\phi$$

$$V = \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} d\phi \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} \sin\theta d\theta \int_{r=0}^{2} r^2 dr$$

$$V = \phi|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cdot -\cos\theta \,|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cdot \frac{r^3}{3}|_0^2$$

$$V = (\frac{\pi}{2} - \frac{\pi}{4}) \cdot (-\cos\frac{\pi}{2} - \left(-\cos\frac{\pi}{4}\right)) \cdot (\frac{8}{3} - \frac{0}{3})$$

$$V = \left(\frac{\pi}{4}\right) \cdot \left(\frac{\sqrt{2}}{2}\right) \cdot \left(\frac{8}{3}\right)$$

$$V = \frac{\pi\sqrt{2}}{3}$$

$$V \approx 1.4810$$

$\therefore$ *the enclosed volume of the closed surface is* $\frac{\pi\sqrt{2}}{3}$ *or approximately* 1.4810 *which was close to the result the MATLAB code generated verifying our results. The MATLAB code generated an answer of* 1.472765 *which has a* 0.56% *difference. To generate a closer result, the discretization step size for the for loops in the MATLAB code would simply be increased which would increase accuracy while also increasing computational expense*

2) Find the area of the closed surface:

$$0 \le r \le 2, \frac{\pi}{4}(45°) \le \phi \le \frac{\pi}{2}(90°), \frac{\pi}{4}(45°) \le \theta \le \frac{\pi}{2}(90°)$$

Finding S_1 $(r = 2)$:

$$dA = r^2 \sin\theta \, d\theta d\phi = 4\sin\theta \, d\theta d\phi$$

$$S_1 = \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} dA$$

$$S_1 = \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} 4\sin\theta \, d\theta d\phi$$

$$S_1 = 4\int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} \sin\theta \, d\theta d\phi$$

$$S_1 = 4\int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} d\phi \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} \sin\theta \, d\theta$$

$$S_1 = 4 \cdot \phi|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cdot -\cos\theta \,|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$S_1 = 4 \cdot \left(\frac{\pi}{4}\right) \cdot \left(\frac{\sqrt{2}}{2}\right)$$

$$S_1 = \frac{\pi\sqrt{2}}{2}$$

Finding S_2 $(\theta = \frac{\pi}{2})$:

$$dA = r \sin\theta \, dr d\phi = r dr d\phi \; (\sin\frac{\pi}{2} = 1)$$

$$S_2 = \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{r=0}^{2} dA$$

$$S_2 = \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{r=0}^{2} r dr d\phi$$

$$S_2 = \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} d\phi \int_{r=0}^{2} r dr$$

$$S_2 = \phi|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cdot \frac{r^2}{2}|_0^2$$

$$S_2 = \left(\frac{\pi}{4}\right) \cdot 2$$

$$S_2 = \frac{\pi}{2}$$

Finding S_3 ($\theta = \frac{\pi}{4}$):

$$dA = r \sin\theta \, dr d\phi = r \frac{\sqrt{2}}{2} dr d\phi \; (\sin\frac{\pi}{4} = \frac{\sqrt{2}}{2})$$

$$S_3 = \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{r=0}^{2} dA$$

$$S_3 = \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{r=0}^{2} r \frac{\sqrt{2}}{2} dr d\phi$$

$$S_3 = \frac{\sqrt{2}}{2} \int_{\phi=\frac{\pi}{4}}^{\frac{\pi}{2}} d\phi \int_{r=0}^{2} r dr$$

$$S_3 = \frac{\sqrt{2}}{2} \cdot \phi|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cdot \frac{r^2}{2}|_0^2$$

$$S_3 = \frac{\sqrt{2}}{2} \cdot \left(\frac{\pi}{4}\right) \cdot 2$$

$$S_3 = \frac{\pi\sqrt{2}}{4}$$

Finding S_4 and S_5 $(\phi = \frac{\pi}{4}$ and $\phi = \frac{\pi}{2})$:

$$dA = rdrd\theta$$

$$S_4 = \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{r=0}^{2} dA$$

$$S_4 = \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{r=0}^{2} rdrd\theta$$

$$S_4 = \int_{\theta=\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_{r=0}^{2} rdr$$

$$S_4 = \theta|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cdot \frac{r^2}{2}|_0^2$$

$$S_4 = \left(\frac{\pi}{4}\right) \cdot 2$$

$$S_4 = \frac{\pi}{2}$$

$$Both\ \phi = \frac{\pi}{4}\ and\ \phi = \frac{\pi}{2}\ have\ the\ same\ surface\ area\ so\ if\ S_4 = \frac{\pi}{2}\ then\ S_5 = S_4 = \frac{\pi}{2}$$

Find Total Area:

$$S_T = S_1 + S_2 + S_3 + S_4 + S_5$$

$$S_T = \frac{\pi\sqrt{2}}{2} + \frac{\pi}{2} + \frac{\pi\sqrt{2}}{4} + \frac{\pi}{2} + \frac{\pi}{2} = \frac{3\pi(2+\sqrt{2})}{4}$$

$$S_T \approx 8.0446$$

$\therefore$ $the\ area\ of\ the\ closed\ surface\ is\ \frac{3\pi(2+\sqrt{2})}{4}\ or\ approximately\ 8.0446\ which\ was\ close$ $to\ the\ result\ the\ MATLAB\ code\ generated\ verifying\ our\ results. The\ MATLAB\ code\ generated$ $an\ answer\ of\ 8.023935\ which\ has\ a\ 0.26\%\ difference.\ To\ generate\ a\ closer\ result, the$ $discretization\ step\ size\ for\ the\ for\ loops\ in\ the\ MATLAB\ code\ would\ simply\ be\ increased\ which$ $would\ increase\ accuracy\ while\ also\ increasing\ computational\ expense$

MATLAB Output to Verify Answers:

Command Window

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a) The enclosed volume of the closed surface is 1.472765
b) The area of the closed surface is 8.023935
>>
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